

LESSON 24 / 第二十四課

Drought-Induced Tree Mortality: From Global Patterns to Xylem Embolism

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Abstract

摘要

This review moves from biome-scale patterns of drought-associated woody-plant mortality to the xylem and carbohydrate processes measured at death. Across 81 sites and 226 forest species, Choat et al. [1] found that 70% operate with hydraulic safety margins narrower than 1 MPa, and that those margins are largely independent of mean annual precipitation. McDowell et al. [2,3] framed mortality as the joint depletion of water and carbon pools and fluxes under drought, heat and rising vapour-pressure deficit. Adams et al. [4] then showed that, at the moment of drought-induced death, percent loss of xylem conductivity was $\geq 60\%$ in every species assessed, whereas non-structural carbohydrate (NSC) decline was not universal and was more frequent in gymnosperms. The literature has therefore shifted from a binary hydraulic-failure versus carbon-starvation choice toward a coupled account in which hydraulic failure is the more consistent terminal observation [2–4]. Open questions remain at the finest scale: how NSC supply maintains living-cell and hydraulic function, and how biotic agents exploit both depleted accounts.

本綜述由乾旱相關木本植物死亡的生物群系格局，寫到死亡當下測得的木質部與碳水化合物過程。Choat 等 [1] 在 81 個樣點、226 個森林樹種中發現，70% 的水力安全邊際窄於 1 MPa，且該邊際大致獨立於年均降水。McDowell 等 [2,3] 把死亡表述為乾旱、高溫與上升水汽壓差下，水分與碳庫及其通量的共同耗竭。Adams 等 [4] 其後顯示，在乾旱致死當下，所有被評估樹種的木質部導水率損失皆 $\geq 60\%$ ，而非結構性碳水化合物（NSC）下降並非普遍，且在裸子植物中較常見。文獻因此已從「水力衰竭對碳飢餓」的二分，轉向以水力衰竭為較一致終末觀察的耦合賬戶 [2–4]。最細尺度上仍開放：NSC 供應如何維持活細胞與水力功能，以及生物性因子如何利用兩個已耗竭的賬戶。

1. Scope of the review

1. 綜述範圍

The object is drought-associated mortality of woody plants, not urban planting demography and not a regional case file. The organising principle is scale (Fig. 1). Section 2 treats global and biome patterns. Section 3 treats whole-tree strategies and the original two-hypothesis framework. Section 4 treats xylem vulnerability curves and safety margins. Section 5 treats observations made at death. Section 6 records unresolved couplings. Study questions at the end test whether the results, not the rhetoric, have been retained.

對象是木本植物與乾旱相關的死亡，不是城市種植人口學，也不是單一地區檔案。組織原則是尺度（圖 1）。第 2 節處理全球與生物群系格局。第 3 節處理整株策略與最初的雙假說框架。第 4 節處理木質部脆弱性曲線與安全邊際。第 5 節處理死亡當下的觀察。第 6 節記錄尚未解決的耦合。文末習題檢驗保留的是結果，而不是修辭。

Figure 1. Nested scales of the drought-mortality literature.



Figure 1. Nested scales used to organise this review, from biome syntheses to conduit-level measurements.

圖1. 本綜述的嵌套尺度：由生物群系綜述到導管尺度測量。

2. Global and biome-scale patterns

2. 全球與生物群系格局

Drought-associated woody-plant mortality has increased in most regions that hold multi-decadal records and is projected to increase further as drought, heat and vapour-pressure deficit rise together [3]. Allen, Breshears and colleagues documented regional die-off events across continents; the review literature treats those events as the phenomenon that physiology must explain, not as anecdotes [3,5]. The first quantitative, multi-biome constraint on the hydraulic side of that explanation is Choat et al. [1]. Compiling published and unpublished vulnerability data, they reported that 70% of 226 forest species from 81 sites operate with safety margins against injurious embolism of less than 1 MPa. Safety margins were largely independent of mean annual precipitation. Wet forests were not, on this metric, safer than dry ones. Gymnosperms as a group held wider margins than angiosperms: 42% of angiosperms versus 6% of gymnosperms fell in the narrowest classes in their comparison [1].

在擁有數十年紀錄的大多數地區，與乾旱相關的木本植物死亡已經增加；隨乾旱、高溫與水汽壓差一同上昇，預計會再增加 [3]。Allen、Breshears 及其同事記錄了跨大陸的區域枯死事件；綜述文獻把這些事件當成生理學必須解釋的現象，而不是軼事 [3,5]。對該解釋在水力一側的第一項多生物群系定量約束，是 Choat 等 [1]。他們彙整已發表與未發表的脆弱性數據，報告 81 個樣點、226 個森林樹種中，70% 對致傷栓塞的安全邊際小於 1 MPa。安全邊際大致獨立於年均降水。以這一度量，濕林並不比乾林更安全。裸子植物整體邊際寬於被子植物：在其比較中，42% 的被子植物對比 6% 的裸子植物落在最窄一組 [1]。

That convergence is the macroscopic result. It does not identify the lethal process inside a dying tree. It states that the operating point of forest species, across rainfall regimes, already sits close to a hydraulic injury threshold. Climate projections that raise evaporative demand therefore move a large fraction of the world's forest flora toward that threshold at once [1,3].

該收斂是宏觀結果。它並不指認一株將死之樹內部的致死過程。它陳述的是：跨越降水體制，森林樹種的運轉點已經靠近水力損傷閾值。提高蒸散需求的氣候推估，因此會同時把世界大部森林植物相推向該閾值 [1,3]。

3. Whole-tree frameworks: two hypotheses, then a coupling

3. 整株框架：兩個假說，然後是耦合

McDowell et al. [6] originally posed two physiological routes. Carbon starvation: stomatal closure under chronic drought reduces assimilation while respiration and other carbon-dependent functions continue, until NSC stores cannot meet demand. Hydraulic failure: tension in the xylem exceeds the air-seeding threshold at pit membranes, emboli accumulate, and water transport collapses [6,7]. Isohydic taxa, which close stomata early to defend Ψ , were argued to be more exposed to the carbon route; anisohydric taxa, which keep stomata open and let Ψ fall, to the hydraulic route [6].

McDowell 等 [6] 最初提出兩條生理路徑。碳飢餓：長期乾旱下氣孔關閉減少同化，而呼吸及其他依賴碳的功能仍在，直至 NSC 儲備無法滿足需求。水力衰竭：木質部張力超過紋孔膜的氣種閾值，栓塞累積，水分運輸崩潰 [6,7]。早關氣孔以保衛 Ψ 的等水種類，被論證為更暴露於碳路徑；讓氣孔保持開放、容許 Ψ 下降的不等水種類，則更暴露於水力路徑 [6]。

Subsequent reviews rejected the binary as a complete description. McDowell [2] already wrote that the two processes are coupled. The 2022 synthesis states the current position directly: drought-associated mortality “results from water and carbon depletion and declines in their fluxes relative to demand by living tissues. These pools and fluxes are interdependent and underlay plant defences against biotic agents” [3]. Rising vapour-pressure deficit accelerates depletion of both accounts; elevated CO₂ can mitigate some of the carbon-side impact, with large uncertainty [3]. Figure 3 summarises that coupling. The debate in the literature is no longer which single hypothesis is true. It is how tightly the two accounts are bound, and which observation is reliable at death.

其後綜述拒絕把二分當成完整描述。McDowell [2] 已寫道兩過程是耦合的。2022 年綜述直接陳述現況：與乾旱相關的死亡「源自水分與碳的耗竭，以及它們相對於活組織需求的通量下降。這些庫與通量相互依賴，並構成植物對生物性因子的防禦基礎」[3]。上升的水汽壓差加速兩個賬戶的耗竭；升高的 CO₂ 可緩解碳側部分影響，不確定性很大 [3]。圖 3 概括該耦合。文獻中的辯論不再是哪一個假說為真，而是兩個賬戶綁得多緊，以及死亡當下哪一項觀察可靠。

Figure 3. Coupled water and carbon accounts (not a binary choice).

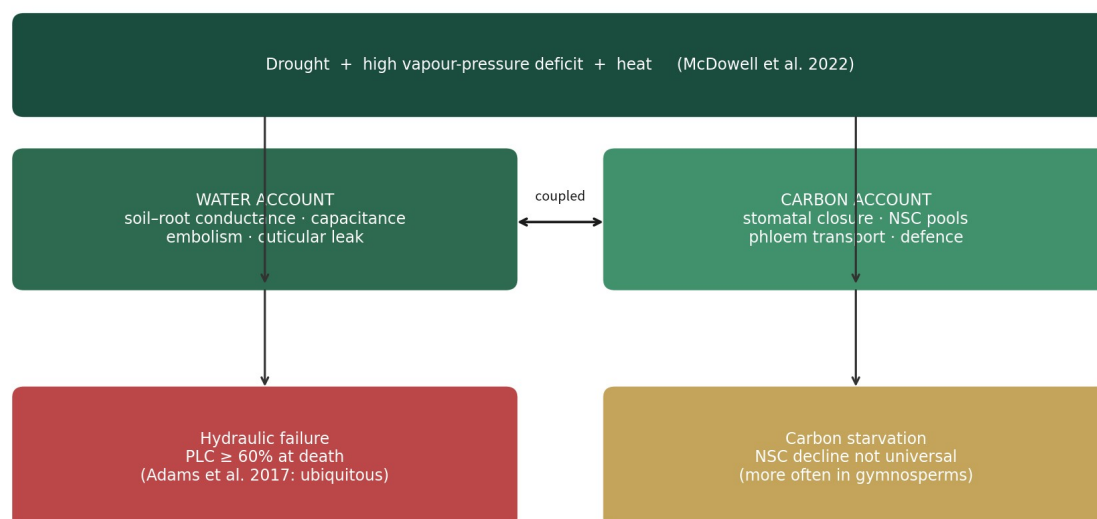


Figure 3. Coupled water and carbon accounts under drought, heat and vapour-pressure deficit, after McDowell et al. [2,3] and Adams et al. [4].

圖 3. 乾旱、高溫與水汽壓差下耦合的水分與碳賬戶，據 McDowell 等 [2,3] 與 Adams 等 [4]。

4. Xylem vulnerability and safety margins

4. 木質部脆弱性與安全邊際

The hydraulic account is quantified on a vulnerability curve: percent loss of conductivity (PLC) as a function of xylem water potential Ψ (Fig. 2). Conventional reference points are P12 (air entry), P50 (50% loss) and P88 (near-complete failure in many angiosperms) [1,8]. A safety margin is the difference between the minimum Ψ experienced in the field ($\Psi_{\min}$) and one of those reference points, most often P50 or P88 [1]. Choat et al. [1] used that construction at global scale. Tyree and Zimmermann [8] remain the anatomical baseline for air seeding at pits; later methodological debate (centrifuge versus optical versus pneumatic techniques) concerns how the curve is measured, not the existence of a threshold [9].

水力賬戶在脆弱性曲線上定量：導水率損失百分率（PLC）作為木質部水勢 Ψ 的函數（圖 2）。慣用參考點是 P12（進氣）、P50（損失 50%）與 P88（許多被子植物接近完全失效）[1,8]。安全邊際是田間經歷的最低 Ψ （ $\Psi_{\min}$ ）與上述參考點之一的差，最常用 P50 或 P88 [1]。Choat 等 [1] 在全球尺度使用該構造。Tyree 與 Zimmermann [8] 仍是紋孔氣種的解剖基線；其後關於離心、光學或氣壓技術的方法辯論，關乎曲線如何測量，而不是閾值是否存在 [9]。

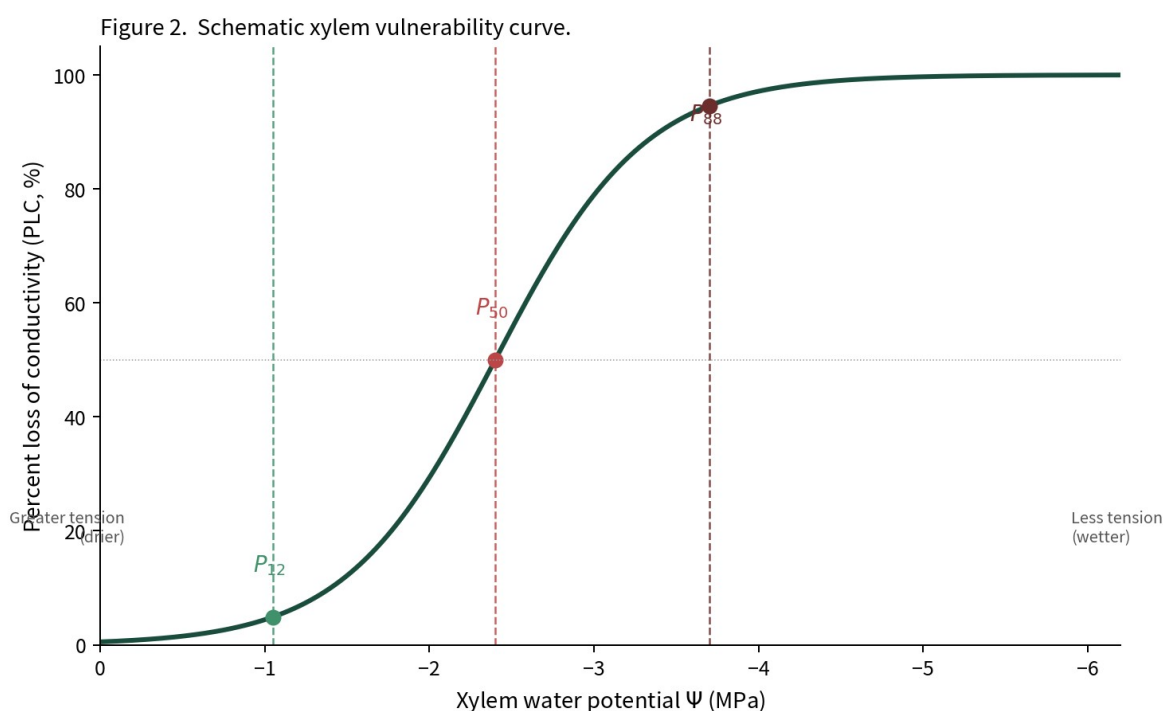


Figure 2. Schematic vulnerability curve. P12, P50 and P88 mark conventional reference points on the PLC– Ψ relationship [1,8].

圖 2. 示意脆弱性曲線。P12、P50 與 P88 標出 PLC– Ψ 關係上的慣用參考點 [1,8]。

Two comparative results follow from the curve. First, angiosperms are on average more vulnerable (less negative P50) than gymnosperms, consistent with the narrower safety margins already noted [1]. Second, because safety margins collapse onto a narrow band across biomes, a given increment of drought intensity is not confined in its risk to arid floras [1]. The curve is the instrument that makes the global pattern in Section 2 a physiological claim rather than a map of dead stands.

由曲線得出兩項比較結果。第一，被子植物平均比裸子植物更脆弱（P50 較不負），與已述較窄安全邊際一致 [1]。第二，因為安全邊際在生物群系間收斂到窄帶，既定的乾旱強度增量，其風險並不限於乾旱植物相 [1]。曲線是使第 2 節全球格局成為生理主張、而不只是枯立木地圖的工具。

5. What is observed at death

5. 死亡當下觀察到什麼

Adams et al. [4] assembled drought-mortality studies under a common physiological frame. Where PLC at death was available, every species had lost at least 60% of xylem hydraulic conductivity, matching proposed survival thresholds. NSC responses at death were diverse. Declines were more common in gymnosperms than in angiosperms, co-varied with hydraulic vulnerability, and were interpreted as possibly contributing to loss of hydraulic function rather than as a standalone lethal depletion [4]. The authors concluded that hydraulic failure at drought-induced mortality was “ubiquitous” in the assembled taxa, and that vegetation models would gain more from hydraulic thresholds than from an unconstrained carbon-starvation switch [4].

Adams 等 [4] 在共同生理框架下彙整乾旱死亡研究。凡有死亡時 PLC 數據的樹種，導水率損失皆至少 60%，與提出的存活閾值相符。死亡時的 NSC 反應多樣。下降在裸子植物中比被子植物更常見，並與水力脆弱性共變，被解釋為可能促成水力功能喪失，而不是單獨的致死耗竭 [4]。作者的結論是：在彙整分類群中，乾旱致死時的水力衰竭是「普遍的」，植被模型從水力閾值得到的增益，會大於一個不受約束的碳飢餓開關 [4]。

That is the microscopic result that currently carries the most cross-species weight. It does not abolish the carbon account. McDowell et al. [3] insist that reduced carbon supply can impair the maintenance of living cells, osmotic function and defence, and can therefore worsen the hydraulic trajectory. The correct reading of Adams et al. [4] beside McDowell et al. [3] is: hydraulic failure is the consistent terminal phenotype; carbon status is a variable co-factor whose weight differs among clades.

這是目前跨物種權重最高的微觀結果。它並不廢除碳賬戶。McDowell 等 [3] 堅持，碳供應減少會損害活細胞維持、滲透功能與防禦，因而可惡化水力軌跡。把 Adams 等 [4] 與 McDowell 等 [3] 放在一起的正確讀法是：水力衰竭是一致的終末表型；碳狀態是權重隨譜系而異的共因子。

6. Unresolved couplings and limits of the evidence

6. 尚未解決的耦合與證據限度

Three limits structure the remaining literature. First, method: vulnerability curves are sensitive to technique, and some published P50 values have been revised [9]. Second, sample: Adams et al. [4] still rest on a modest number of species with simultaneous PLC and NSC at death; gymnosperm–angiosperm differences may shift as the set grows. Third, biotic agents sit at the end of both accounts and are under-integrated in the physiological syntheses, even though defence is named as a carbon- and water-dependent function [3]. The review record does not yet support a single lethal Ψ or a single lethal NSC concentration that can be copied from one species to another.

三項限度構成剩餘文獻的結構。第一，方法：脆弱性曲線對技術敏感，部分已發表 P50 曾被修正 [9]。第二，樣本：Adams 等 [4] 仍建立在死亡時同時具 PLC 與 NSC 的有限樹種上；隨著集合擴大，裸子–被子差異可能移動。第三，生物性因子位於兩個賬戶的末端，在生理綜述中整合不足，即使防禦已被點名為依賴碳與水的功能 [3]。綜述紀錄尚不支持一個可從一樹種複製到另一樹種的單一致死 Ψ 或單一致死 NSC 濃度。

7. Study questions

7. 研習題

1. Choat et al. [1] found that 70% of 226 forest species operate with hydraulic safety margins:

1. Choat 等 [1] 發現 226 個森林樹種中 70% 的水力安全邊際為：

A. Wider than 3 MPa in all wet forests.

A. 所有濕林都寬於 3 MPa。

B. Narrower than 1 MPa.

B. 窄於 1 MPa。

C. Exactly equal to P88.

C. 剛好等於 P88。

D. Proportional to city size.

D. 與城市面積成正比。

2. In Choat et al. [1], safety margins were:

2. 在 Choat 等 [1] 中，安全邊際：

A. Strictly determined by mean annual precipitation.

A. 嚴格由年均降水決定。

B. Largely independent of mean annual precipitation.

B. 大致獨立於年均降水。

C. Undefined for gymnosperms.

C. 對裸子植物無定義。

D. Measured only in cities.

D. 只在城市測量。

3. Adams et al. [4] reported that at drought-induced mortality, PLC was:

3. Adams 等 [4] 報告乾旱致死時 PLC 為：

A. Always zero.

A. 總是為零。

B. $\geq 60\%$ in every species assessed.

B. 所有被評估樹種皆 $\geq 60\%$ 。

C. $\geq 60\%$ only in angiosperms.

C. 只有被子植物 $\geq 60\%$ 。

D. Unrelated to conductivity.

D. 與導水率無關。

4. NSC decline at death in Adams et al. [4] was:

4. Adams 等 [4] 中死亡時的 NSC 下降是：

A. Universal and identical in all clades.

A. 在所有譜系普遍且相同。

B. Diverse, and more common in gymnosperms.

B. 多樣，且在裸子植物中較常見。

C. Proof that carbon starvation is the sole cause.

C. 證明碳飢餓是唯一原因。

D. Never observed.

D. 從未被觀察。

5. McDowell et al. [3] treat water and carbon pools as:

5. McDowell 等 [3] 把水分與碳庫視為：

A. Independent switches.

A. 互相獨立的開關。

B. Interdependent pools and fluxes that also underlie defence.

B. 相互依賴的庫與通量，亦構成防禦基礎。

C. Irrelevant to VPD.

C. 與 VPD 無關。

D. Nursery grading criteria.

D. 苗圃分級標準。

6. P50 on a vulnerability curve is:

6. 脆弱性曲線上的 P50 是：

A. The Ψ at 50% loss of conductivity.

A. 導水率損失 50% 時的 Ψ 。

B. Fifty years of tree age.

B. 樹齡五十年。

C. A soil bulk-density threshold.

C. 土壤容重閾值。

D. Mean annual precipitation.

D. 年均降水。

7. Gymnosperms in Choat et al. [1] generally showed:

7. Choat 等 [1] 中裸子植物一般表現為：

A. Narrower safety margins than angiosperms.

A. 安全邊際窄於被子植物。

B. Wider safety margins than angiosperms.

B. 安全邊際寬於被子植物。

C. No xylem.

C. 無木質部。

D. Instant leaf-off.

D. 立即落葉。

8. The current synthesis position on the two original hypotheses is that:

8. 對最初兩個假說，目前綜述立場是：

A. Only carbon starvation is true.

A. 只有碳飢餓為真。

B. Only hydraulic failure is ever discussed.

B. 只討論水力衰竭。

C. The processes are coupled; hydraulic failure is the more consistent observation at death.

C. 過程耦合；水力衰竭是死亡時較一致的觀察。

D. Neither process exists.

D. 兩過程都不存在。

9. Rising vapour-pressure deficit in McDowell et al. [3] is treated as:

9. McDowell 等 [3] 把上升的水汽壓差視為：

A. A factor that accelerates depletion of water and carbon accounts.

A. 加速水分與碳賬戶耗竭的因子。

B. Irrelevant in forests.

B. 在森林中無關。

C. A substitute for P50.

C. P50 的替代。

D. A pruning specification.

D. 修剪規格。

10. A species-specific lethal Ψ that can be copied across the world's flora:

10. 一個可複製到全球植物相的物種特異致死 Ψ ：

A. Is provided as a single number in Choat et al. [1].

A. Choat 等 [1] 以單一數字提供。

B. Is not supported by the present review record.

B. 現有綜述紀錄並不支持。

C. Equals -1.0 MPa for all angiosperms.

C. 所有被子植物等於 -1.0 MPa。

D. Equals zero NSC.

D. 等於零 NSC。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-A 7-B 8-C 9-A 10-B

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